

**Michigan
PROCEDURE**
**LEFT VENTRICULAR ASSIST DEVICE
(LVAD)**

Initial Date:

Revised Date: 01/27/2023

Section 7-28

Left Ventricular Assist Device

A Left Ventricular Assist Device (LVAD) is an implanted device that pumps blood from the left ventricle into the aorta to support circulation. For some of these patients this device is a bridge to transplant but for others it is a life prolonging therapy if transplant is not an option. Care of patients supported by these devices can present a challenge for care givers in the pre-hospital environment. This document provides guidance for the provision of emergency care for patients in the pre-hospital environment who have an LVAD in place. Contact VAD coordinator/center for devices which you are unfamiliar with or require assistance with.

Contact Information:

Program Name:

Phone: _

Request VAD Coordinator and state patient's name

VAD Pager number:

Contact Information:

Program Name:

Phone: _

Request VAD Coordinator and state patient's name

VAD Pager number:

1. LVAD's create non-pulsatile flow; it may be difficult to obtain vital signs using standard equipment and or methods. Utilize skin color, mental status and capillary refill to assess the patient.
2. The device supports left ventricular function and is dependent on some right heart function and adequate circulating volume. Even minor volume depletion may cause diminished perfusion and require fluid administration.
3. All LVAD patients are anticoagulated.
4. LVAD's are powered electrically, a driveline exits the body, connects to a "controller" which in turn is connected to a power source. Proper functioning of the device is dependent on the integrity of these connections. Exercise caution related to the drive line, which exits through the skin in the upper abdomen. Do not cut, pull or damage it in any way. It will be secured by some type of binder or other device to protect it.
5. Connections should not be forced together or apart. All connections are secured by a locking device.
6. Generally, patients, their families and caregivers are familiar with the operation of the device and should accompany the patient as a resource for operation of the device if promptly available.
7. All LVAD patients are assigned a hospital-based coordinator who is available by phone and should be contacted urgently.

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8. All LVAD patients should have a “go bag” close by which contains an additional power supply as well as an extra controller. This should be brought with the patient to the hospital. This should contain charged batteries, a back-up controller and a power-based unit.
9. If possible, the patient should be transported with four fully charged batteries. Two will be connected to the patient and the other will serve as backups.
10. Most issues will be the result of medical problems rather than device failure.

Procedure

Do NOT use the following devices on an LVAD patient

- AED
- Mechanical Compression Device

1. Assess the patient for signs of life and function of the device
 - A. Awake and or alert
 - B. Satisfactory capillary refill
 - C. Audible whine/hum in the region around the heart and or left upper abdomen
 - D. Check all connections, tighten as indicated to be sure they are secure
 - E. Identify any alarms that are heard or visible on controller and relay information to VAD coordinator.
 - F. If able, begin to assemble components or have the patient’s designated LVAD companion gather components that will accompany patient
 - a. Extra controller
 - b. Extra batteries
 - c. Power unit (charger) and or A/C adapter
2. Assess for other medical issues
 -  A. Start an IV and a fluid bolus if volume depletion is felt to be present
 - B. Control bleeding
 -  C. Attach monitor and assess rhythm
 - a. LVAD patients may have life threatening arrhythmias at baseline including VF or VT. Ask the patient, companion, or LVAD coordinator what the patient’s baseline rhythm is.
 - b. If the patient is unstable and they are in an arrhythmia that is not their baseline treat the arrhythmia
 - c. Defibrillation, cardioversion, and external pacing are allowed if indicated. You do not need to disconnect the device.
 - D. Follow appropriate medical protocol
 - E. CPR compressions should only be performed as a last resort.
 -  a. Consult with Medical Control immediately if the device is non-functioning and you are starting CPR.
 - F. Prepare for transport to MCA approved LVAD hospital

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3. Consult with LVAD coordinator
 - A. Patient or companion should have emergency contact information
 - B. Report information from the controller including any alarms
 - C. Change battery or power source as requested
 - D. Change controller as requested-be sure patient is laying or sitting down as pump will stop briefly
 4. Transport to an MCA approved LVAD Center
 - A.
 - B.
 - C.